

Introduction to Machine Learning

Week 7

Prof. B. Ravindran, IIT Madras

1. (1 Mark) Define active learning:

- (a) A learning approach where the algorithm passively receives all training data at once
- (b) A technique where the model learns from its own predictions without human intervention
- (c) An iterative learning process where the model selects the most informative data points for labeling
- (d) A method where the model randomly selects data points for training to reduce bias

Sol. (c) - Refer to the lectures

2. (2 Mark) Given 100 distinct data points, if you sample 100 times with replacement, what is the expected number of distinct points you will obtain?

- (a) Approximately 50
- (b) Approximately 63
- (c) Exactly 100
- (d) Approximately 37

Sol. (b) -

Probability of not selecting a point in 100 tries = $\frac{99}{100}^{100}$

Probability of selecting a point atleast once in 100 tries = $1 - \frac{99}{100}^{100}$

Expectation = $\sum x \times P(x) = 100(1 - (\frac{99}{100})^{100}) \approx 63$

3. (1 Mark) What is the key difference between bootstrapping and cross-validation?

- (a) Bootstrapping uses the entire dataset for training, while cross-validation splits the data into subsets
- (b) Cross-validation allows replacement, while bootstrapping does not
- (c) Bootstrapping creates multiple samples with replacement, while cross-validation creates subsets without replacement
- (d) Cross-validation is used for model selection, while bootstrapping is only used for uncertainty estimation

Sol. (c) - Refer to the lectures

4. (2 Marks) Consider the following confusion matrix for a binary classification problem:

	Predicted Positive	Predicted Negative
Actual Positive	85	15
Actual Negative	20	80

What are the precision, recall, and accuracy of this classifier?

- (a) Precision: 0.81, Recall: 0.85, Accuracy: 0.83
- (b) Precision: 0.85, Recall: 0.81, Accuracy: 0.85
- (c) Precision: 0.80, Recall: 0.85, Accuracy: 0.82
- (d) Precision: 0.85, Recall: 0.85, Accuracy: 0.80

Sol. (a) -

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{85}{85+20} = \frac{85}{105} \approx 0.81$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{85}{85+15} = \frac{85}{100} = 0.85$$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{85+80}{200} = \frac{165}{200} = 0.825 \approx 0.83$$

5. (1 Mark) AUC for your newly trained model is 0.5. Is your model prediction completely random?
- (a) Yes
 - (b) No
 - (c) ROC curve is needed to derive this conclusion
 - (d) Cannot be determined even with ROC

Sol. (c) - An AUC of 0.5 suggests that the model is either making random predictions, predicting all instances as a single class, or making systematically incorrect predictions that could be corrected by inverting its outputs.

6. (1 Mark) You are building a model to detect cancer. Which metric will you prefer for evaluating your model?
- (a) Accuracy
 - (b) Sensitivity
 - (c) Specificity
 - (d) MSE

Sol. (b) - In medical application, FP is the most important (which sensitivity captures)

7. (1 Mark) You have 2 binary classifiers A and B. A has accuracy=0% and B has accuracy=50%. Which classifier is more useful?
- (a) A
 - (b) B
 - (c) Both are good
 - (d) Cannot say

Sol. (a) - Flip the labels and get 100% accuracy!

8. (1 Mark) You have a special case where your data has 10 classes and is sorted according to target labels. You attempt 5-fold cross validation by selecting the folds sequentially. What can you say about your resulting model?

- (a) It will have 100% accuracy.
- (b) It will have 0% accuracy.
- (c) It will have close to perfect accuracy.
- (d) Accuracy will depend on the compute power available for training.

Sol. (b) - The training and test sets are partitioned in a way that some classes are only present in the test set. This means the classifier will never learn about these classes and therefore cannot predict them